

OUTSIDE OPTIONS AND RISK ATTITUDE

Gregorio Curello
Mannheim

Ludvig Sinander
Oxford

Mark Whitmeyer
Arizona State

29 September 2026

paper: [arXiv.org/abs/2509.14732](https://arxiv.org/abs/2509.14732)

Motivation

(Effective) risk attitude: how a decision-maker chooses
among risky prospects (=lotteries).

Motivation

(Effective) risk attitude: how a decision-maker chooses
among risky prospects (=lotteries).

A key determinant of many economic phenomena.

Examples:

prospects

projects

Motivation

(Effective) risk attitude: how a decision-maker chooses
among risky prospects (=lotteries).

A key determinant of many economic phenomena.

Examples:

prospects

projects

health-insurance plans

Motivation

(Effective) risk attitude: how a decision-maker chooses
among risky prospects (=lotteries).

A key determinant of many economic phenomena.

Examples:

prospects

projects

health-insurance plans

occupations

spouses

locations

Motivation

(Effective) risk attitude: how a decision-maker chooses
among risky prospects (=lotteries).

A key determinant of many economic phenomena.

In most economic contexts, chosen prospect is not all there is:

prospects	outside option	
projects	job loss	(for manager)
health-insurance plans		
occupations		
spouses		
locations		

Motivation

(Effective) risk attitude: how a decision-maker chooses
among risky prospects (=lotteries).

A key determinant of many economic phenomena.

In most economic contexts, chosen prospect is not all there is:

prospects	outside option
projects	job loss (for manager)
health-insurance plans	public option
occupations	
spouses	
locations	

Motivation

(Effective) risk attitude: how a decision-maker chooses
among risky prospects (=lotteries).

A key determinant of many economic phenomena.

In most economic contexts, chosen prospect is not all there is:

prospects	outside option
projects	job loss (for manager)
health-insurance plans	public option
occupations	retraining
spouses	divorce
locations	bailout (e.g. FEMA)

Questions

Framework:

$$\text{effective risk att.} = \text{'true' risk att.} + \text{effect of o.o.}$$

Questions

Framework:

$$\underbrace{\text{effective risk att.}}_{\text{observable}} = \underbrace{\text{'true' risk att.}}_{\text{unobservable}} + \underbrace{\text{effect of o.o.}}_{\text{unobservable}}$$

Identification: what do choices reveal about 'true' risk attitude?

- important for policy / welfare analysis

Questions

Framework:

$$\underbrace{\text{effective risk att.}}_{\text{observable}} = \underbrace{\text{'true' risk att.}}_{\text{unobservable}} + \underbrace{\text{effect of o.o.}}_{\text{unobservable}}$$

Identification: what do choices reveal about 'true' risk attitude?

- important for policy / welfare analysis
- Thm 1(i): 'true' is more risk-averse than effective.

Questions

Framework:

$$\underbrace{\text{effective risk att.}}_{\text{observable}} = \underbrace{\text{'true' risk att.}}_{\text{unobservable}} + \underbrace{\text{effect of o.o.}}_{\text{unobservable}}$$

Identification: what do choices reveal about 'true' risk attitude?

- important for policy / welfare analysis
- Thm 1(i): 'true' is more risk-averse than effective.
- Thm 1(ii): nothing more can be learned.

Questions

Framework:

$$\text{effective risk att.} = \text{'true' risk att.} + \text{effect of o.o.}$$

Identification: what do choices reveal about 'true' risk attitude?

- important for policy / welfare analysis
- Thm 1(i): 'true' is more risk-averse than effective.
- Thm 1(ii): nothing more can be learned.

Comparative statics: how does effective risk att. vary
with 'true' risk att. & o.o.? (Th'm 2)

Questions

Framework:

$$\text{effective risk att.} = \text{'true' risk att.} + \text{effect of o.o.}$$

Identification: what do choices reveal about 'true' risk attitude?

- important for policy / welfare analysis
- Thm 1(i): 'true' is more risk-averse than effective.
- Thm 1(ii): nothing more can be learned.

Comparative statics: how does effective risk att. vary
with 'true' risk att. & o.o.? (Th'm 2)

Specialness: which transform'ns of a decision problem
necessarily reduce risk-aversion?

- adding an o.o.: yes by Th'm 1(i)

Questions

Framework:

$$\text{effective risk att.} = \text{'true' risk att.} + \text{effect of o.o.}$$

Identification: what do choices reveal about 'true' risk attitude?

- important for policy / welfare analysis
- Thm 1(i): 'true' is more risk-averse than effective.
- Thm 1(ii): nothing more can be learned.

Comparative statics: how does effective risk att. vary
with 'true' risk att. & o.o.? (Th'm 2)

Specialness: which transform'ns of a decision problem
necessarily reduce risk-aversion?

- adding an o.o.: yes by Th'm 1(i)
- any other transform'n: no! (Th'm 3)

Literature

Th'm 1(i) formalises idea that o.o. increases risk appetite.

An old idea. For example, Adam Smith (1776) argued that limited liability (o.o. = bankruptcy) increases risk-taking.

Details: slide 31. See also e.g. Jensen and Meckling (1976, section 4.1),
Golbe (1981, 1988),
Gollier, Koehl and Rochet (1997).

An important idea, e.g. for financial stability.

Literature

Th'm 1(i) formalises idea that o.o. increases risk appetite.

An old idea. For example, Adam Smith (1776) argued that limited liability (o.o. = bankruptcy) increases risk-taking.

Details: slide 31. See also e.g. Jensen and Meckling (1976, section 4.1),
Golbe (1981, 1988),
Gollier, Koehl and Rochet (1997).

An important idea, e.g. for financial stability.

Th'ns 1(ii), 2, 3: no close parallels that we know of.

Broader literature

Various literatures recognise that

$$\text{effective risk att.} = \text{'true' risk att.} + \text{economic forces}$$

Broader literature

Various literatures recognise that

$$\text{effective risk att.} = \text{'true' risk att.} + \text{economic forces}$$

Some economic forces that have been studied:

- background risk.....papers: slide 34
- contracts
 - employment.....papers: slide 35
 - financing papers: slide 36
- having an audience
 - career concerns.....papers: slide 37
 - disclosure..... Ben-Porath, Dekel and Lipman, 2018
- (in)flexibility papers: slide 38
- competition.....papers: slides 39–42

Plan

Setup and background

The outside-option model

How an o.o. shapes risk attitude (Proposition 1)

Identification (Theorem 1)

Comparative statics (Theorem 2)

What is special about o.o.? (Theorem 3)

Plan

Setup and background

The outside-option model

How an o.o. shapes risk attitude (Proposition 1)

Identification (Theorem 1)

Comparative statics (Theorem 2)

What is special about o.o.? (Theorem 3)

Setup

Alternatives: $x, y \in X \neq \emptyset$.

Risky prospects / simple lotteries: $p, q \in \Delta^0(X)$

$$:= \left\{ p : X \rightarrow [0, 1] : |\text{supp}(p)| < \infty \ \& \ \sum_{x \in \text{supp}(p)} p(x) = 1 \right\}$$

Preferences $\succeq, \succeq'$: complete transitive relations on $\Delta^0(X)$.

Setup

Alternatives: $x, y \in X \neq \emptyset$.

Risky prospects / simple lotteries: $p, q \in \Delta^0(X)$

$$:= \left\{ p : X \rightarrow [0, 1] : |\text{supp}(p)| < \infty \ \& \ \sum_{x \in \text{supp}(p)} p(x) = 1 \right\}$$

Preferences $\succeq, \succeq'$: complete transitive relations on $\Delta^0(X)$.

$\succeq$ is EU iff $\exists u : X \rightarrow \mathbf{R}$ s.t. $p \succeq q \iff \int u dp \geq \int u dq$.

$\hookrightarrow$ the function u is called risk attitude

$\hookrightarrow \alpha u + \beta$ is identified off $\succeq$ ($\alpha > 0$ & $\beta \in \mathbf{R}$ unknown)

Setup

Alternatives: $x, y \in X \neq \emptyset$.

Risky prospects / simple lotteries: $p, q \in \Delta^0(X)$

$$:= \left\{ p : X \rightarrow [0, 1] : |\text{supp}(p)| < \infty \ \& \ \sum_{x \in \text{supp}(p)} p(x) = 1 \right\}$$

Preferences $\succeq, \succeq'$: complete transitive relations on $\Delta^0(X)$.

$\succeq$ is EU iff $\exists u : X \rightarrow \mathbf{R}$ s.t. $p \succeq q \iff \int u dp \geq \int u dq$.

$\hookrightarrow$ the function u is called risk attitude

$\hookrightarrow \alpha u + \beta$ is identified off $\succeq$ ($\alpha > 0$ & $\beta \in \mathbf{R}$ unknown)

$\succeq$ is less risk-averse than $\succeq'$ iff $\forall x \in X$ & $\forall p \in \Delta^0(X)$,

$$x \succeq(\succ) p \implies x \succeq'(\succ') p.$$

Setup

Alternatives: $x, y \in X \neq \emptyset$.

Risky prospects / simple lotteries: $p, q \in \Delta^0(X)$

$$:= \left\{ p : X \rightarrow [0, 1] : |\text{supp}(p)| < \infty \ \& \ \sum_{x \in \text{supp}(p)} p(x) = 1 \right\}$$

Preferences $\succeq, \succeq'$: complete transitive relations on $\Delta^0(X)$.

$\succeq$ is EU iff $\exists u : X \rightarrow \mathbf{R}$ s.t. $p \succeq q \iff \int u dp \geq \int u dq$.

$\hookrightarrow$ the function u is called risk attitude

$\hookrightarrow \alpha u + \beta$ is identified off $\succeq$ ($\alpha > 0$ & $\beta \in \mathbf{R}$ unknown)

u is less risk-averse than v iff $\forall x \in X$ & $\forall p \in \Delta^0(X)$,

$$u(x) \geq (>) \int u dp \implies v(x) \geq (>) \int v dp.$$

Pratt's theorem (part 1)

For $X \neq \emptyset$ and $u, v : X \rightarrow \mathbf{R}$, the following are equivalent:

(A) u is less risk-averse than v .

Pratt's theorem (part 1)

For $X \neq \emptyset$ and $u, v : X \rightarrow \mathbf{R}$, the following are equivalent:

(A) u is less risk-averse than v .

(B) $\exists$ increasing convex $\phi : \text{co}(v(X)) \rightarrow \mathbf{R}$
that is strictly increasing on $v(X)$ & satisfies $u = \phi \circ v$.

Pratt's theorem (part 1)

For $X \neq \emptyset$ and $u, v : X \rightarrow \mathbf{R}$, the following are equivalent:

(A) u is less risk-averse than v .

(B) $\exists$ increasing convex $\phi : \text{co}(v(X)) \rightarrow \mathbf{R}$
that is strictly increasing on $v(X)$ & satisfies $u = \phi \circ v$.

(C) The following two properties hold:

(I) $\forall x, y \in X, u(x) \geq (>) u(y) \implies v(x) \geq (>) v(y)$.

(II) $\forall x, y, z \in X$, if $u(x) < u(y) < u(z)$, then

$$\frac{u(z) - u(y)}{u(y) - u(x)} \geq \frac{v(z) - v(y)}{v(y) - v(x)}.$$

Pratt's theorem (part 2)

If in addition X is an open convex subset of $\mathbf{R}$
& u, v are C^2 with $u' > 0 < v'$, then
 u is less risk-averse than v iff $u''/u' \geq v''/v'$.

Plan

Setup and background

The outside-option model

How an o.o. shapes risk attitude (Proposition 1)

Identification (Theorem 1)

Comparative statics (Theorem 2)

What is special about o.o.? (Theorem 3)

The outside-option model

Decision-maker has

- EU risk attitude $v : X \rightarrow \mathbf{R}$
- outside option worth $\sim F$.

The outside-option model

Decision-maker has

- EU risk attitude $v : X \rightarrow \mathbf{R}$
- outside option worth $\sim F$.

o.o. may be unavailable, i.e. ‘worth $-\infty$ ’

$\implies F : \mathbf{R} \rightarrow [0, 1]$ is an extended CDF. (def’n: slide 32)

The outside-option model

Decision-maker has

- EU risk attitude $v : X \rightarrow \mathbf{R}$
- outside option worth $\sim F$.

o.o. may be unavailable, i.e. ‘worth $-\infty$ ’

$\implies F : \mathbf{R} \rightarrow [0, 1]$ is an extended CDF. (def’n: slide 32)

Decision-maker’s valuation of alternative $x \in X$ if o.o. worth k :

$$\max\{v(x), k\}$$

(Compare realised o.o. with x ; if better, exercise.)

The outside-option model

Decision-maker has

- EU risk attitude $v : X \rightarrow \mathbf{R}$
- outside option worth $\sim F$.

o.o. may be unavailable, i.e. ‘worth $-\infty$ ’

$\implies F : \mathbf{R} \rightarrow [0, 1]$ is an extended CDF. (def’n: slide 32)

Decision-maker’s valuation of alternative $x \in X$:

$$\int \max\{v(x), k\} F(\mathrm{d}k) = F(v(x))v(x) + \int_{(v(x), +\infty)} k F(\mathrm{d}k).$$

(Compare realised o.o. with x ; if better, exercise.)

Where do outside options come from?

Where do outside options come from?

Physical o.o. $\sim \mu \in \Delta(Y \cup \{\emptyset\})$,

payoff fn $w : Y \rightarrow \mathbf{R}$

‘ $\emptyset$ ’ means ‘unavailable’

convention: $w(\emptyset) := -\infty$

Where do outside options come from?

Physical o.o. $\sim \mu \in \Delta(Y \cup \{\emptyset\})$, ‘ $\emptyset$ ’ means ‘unavailable’
payoff fn $w : Y \rightarrow \mathbf{R}$ convention: $w(\emptyset) := -\infty$

$$\implies F(k) = \mu(\{y \in Y \cup \{\emptyset\} : w(y) \leq k\}) \quad \forall k \in \mathbf{R}. \quad (\star)$$

Where do outside options come from?

Physical o.o. $\sim \mu \in \Delta(Y \cup \{\emptyset\})$, ‘ $\emptyset$ ’ means ‘unavailable’
payoff fn $w : Y \rightarrow \mathbf{R}$ convention: $w(\emptyset) := -\infty$
 $\implies F(k) = \mu(\{y \in Y \cup \{\emptyset\} : w(y) \leq k\}) \quad \forall k \in \mathbf{R}. \quad (\star)$

Special case: X -valued o.o., i.e. $Y = X$ & $w = v$.

Where do outside options come from?

Physical o.o. $\sim \mu \in \Delta(Y \cup \{\emptyset\})$, ‘ $\emptyset$ ’ means ‘unavailable’
payoff fn $w : Y \rightarrow \mathbf{R}$ convention: $w(\emptyset) := -\infty$
 $\implies F(k) = \mu(\{y \in Y \cup \{\emptyset\} : w(y) \leq k\}) \quad \forall k \in \mathbf{R}. \quad (\star)$

Special case: X -valued o.o., i.e. $Y = X$ & $w = v$.

- Implies F concentrated on $v(X) \cup \{-\infty\}$.

Where do outside options come from?

Physical o.o. $\sim \mu \in \Delta(Y \cup \{\emptyset\})$, ‘ $\emptyset$ ’ means ‘unavailable’
payoff fn $w : Y \rightarrow \mathbf{R}$ convention: $w(\emptyset) := -\infty$
 $\implies F(k) = \mu(\{y \in Y \cup \{\emptyset\} : w(y) \leq k\}) \quad \forall k \in \mathbf{R}. \quad (\star)$

Special case: X -valued o.o., i.e. $Y = X$ & $w = v$.

- Implies F concentrated on $v(X) \cup \{-\infty\}$.
- Conversely, any ECDF F concentrated on $v(X) \cup \{-\infty\}$ arises via $(\star)$ from some X -valued o.o. $\mu \in \Delta(X \cup \{-\infty\})$.
- (Detail: above makes sense provided $v(X)$ is Borel.)

(skip to slide 23)

Behavioural implications

Say that (v, F) is o.o. representation of $\succsim$ iff

$$p \succsim q \iff \left\{ \begin{array}{l} \int \left(\int \max\{v(x), k\} F(\mathrm{d}k) \right) p(\mathrm{d}x) \\ \geq \int \left(\int \max\{v(x), k\} F(\mathrm{d}k) \right) q(\mathrm{d}x). \end{array} \right.$$

Behavioural implications

Say that (v, F) is o.o. representation of $\succeq$ iff

$$p \succeq q \iff \left\{ \begin{array}{l} \underbrace{\int \left(\int \max\{v(x), k\} F(dk) \right) p(dx)}_{=\alpha u(x) + \beta} \\ \geq \int \left(\int \max\{v(x), k\} F(dk) \right) q(dx). \end{array} \right.$$

Note: (v, F) o.o. rep'n of $\succeq$

$\implies \succeq$ is EU with risk att. u where $\exists \alpha > 0$ & $\beta \in \mathbf{R}$ s.t.

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(dk) \quad \forall x \in X.$$

$\hookrightarrow u$ is decision-maker's effective risk attitude.

Behavioural implications

Say that (v, F) is o.o. representation of $\succeq$ iff

$$p \succeq q \iff \left\{ \begin{array}{l} \underbrace{\int \left(\int \max\{v(x), k\} F(dk) \right) p(dx)}_{=\alpha u(x) + \beta} \\ \geq \int \left(\int \max\{v(x), k\} F(dk) \right) q(dx). \end{array} \right.$$

Note: (v, F) o.o. rep'n of $\succeq$

$\implies \succeq$ is EU with risk att. u where $\exists \alpha > 0$ & $\beta \in \mathbf{R}$ s.t.

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(dk) \quad \forall x \in X.$$

$\hookrightarrow u$ is decision-maker's effective risk attitude.

Corollary: $\succeq$ admits an o.o. representation iff $\succeq$ is EU.

Plan

Setup and background

The outside-option model

How an o.o. shapes risk attitude (Proposition 1)

Identification (Theorem 1)

Comparative statics (Theorem 2)

What is special about o.o.? (Theorem 3)

How an o.o. shapes risk attitude

Question: how does shape of o.o. dist'n F shape wedge between effective risk att. u from 'true' risk att. v ?

How an o.o. shapes risk attitude

Proposition 1. For $X \neq \emptyset$,
 $u, v : X \rightarrow \mathbf{R}$,
& ECDF F ,

the following are equivalent:

(i) $\exists \alpha > 0$ & $\beta \in \mathbf{R}$ such that

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(\mathrm{d}k) \quad \forall x \in X.$$

(ii) $\int_{(0,+\infty)} k F(\mathrm{d}k) < +\infty$, and

$$\exists \phi : \text{co}(v(X)) \rightarrow \mathbf{R} \quad \text{s.t.} \quad u = \phi \circ v,$$

$$\phi \text{ is abs. cont's} \quad \& \quad \phi' \propto F.$$

How an o.o. shapes risk attitude: **smoothish** case

Proposition 1. For $X \neq \emptyset$,

$u, v : X \rightarrow \mathbf{R}$,

& ECDF F that is C^1 with $F > 0$ on $(\inf v(X), +\infty)$,

the following are equivalent:

(i) $\exists \alpha > 0$ & $\beta \in \mathbf{R}$ such that

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(\mathrm{d}k) \quad \forall x \in X.$$

(ii') $\int_{(0, +\infty)} k F(\mathrm{d}k) < +\infty$, and

$\exists \phi : \text{co}(v(X)) \rightarrow \mathbf{R}$ s.t. $u = \phi \circ v$,

ϕ is C^2 & $\phi''/\phi' = F'/F$.

How an o.o. shapes risk attitude: **monetary** case

Proposition 1. For $X \neq \emptyset$ open convex subset of $\mathbf{R}$,
 $u, v : X \rightarrow \mathbf{R}$ that are C^2 with $u' > 0 < v'$,
& ECDF G that is C^1 with $G > 0$ on $(\inf X, +\infty)$
& concentrated on $X \cup \{-\infty\}$,
the following are equivalent:

(i) $\exists \alpha > 0$ & $\beta \in \mathbf{R}$ such that

$$\alpha u(x) + \beta = \int \max\{v(x), v(y)\} G(dy) \quad \forall x \in X.$$

(ii'') $\int_{(0, +\infty)} v dG < +\infty$, and

$$\frac{u''}{u'} = \frac{v''}{v'} + \frac{G'}{G}.$$

$G \equiv F \circ v$ is dist'n of o.o. denominated in dollars (not utils).

Sketch proof of Proposition 1

(i) $\exists \alpha > 0, \beta \in \mathbf{R}$ such that

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(\mathrm{d}k) \quad \forall x \in X.$$

Sketch proof of Proposition 1

Note: (i) is equivalent (by change of var'ble $\ell := v(x)$) to:

$\exists \alpha > 0, \beta \in \mathbf{R} \ \& \ \phi : \text{co}(v(X)) \rightarrow \mathbf{R}$ such that

$u = \phi \circ v$ and

$$\alpha \phi(\ell) + \beta = \int \max\{\ell, k\} F(\mathrm{d}k) \quad \forall \ell \in \text{co}(v(X)).$$

Sketch proof of Proposition 1

Note: (i) is equivalent (by change of var'ble $\ell := v(x)$) to:

$\exists \alpha > 0, \beta \in \mathbf{R} \ \& \ \phi : \text{co}(v(X)) \rightarrow \mathbf{R}$ such that

$u = \phi \circ v$ and

$$\alpha \phi(\ell) + \beta = \int \max\{\ell, k\} F(\mathrm{d}k) \quad \forall \ell \in \text{co}(v(X)).$$

Note: $\frac{\mathrm{d}}{\mathrm{d}\ell} \text{RHS} = \int \mathbf{1}_{k \leq \ell} F(\mathrm{d}k) = \int_{-\infty}^{\ell} \mathrm{d}F = F(\ell).$

Sketch proof of Proposition 1

Note: (i) is equivalent (by change of var'ble $\ell := v(x)$) to:

$\exists \alpha > 0, \beta \in \mathbf{R} \ \& \ \phi : \text{co}(v(X)) \rightarrow \mathbf{R}$ such that

$u = \phi \circ v$ and

$$\alpha \phi(\ell) + \beta = \int \max\{\ell, k\} F(\mathrm{d}k) \quad \forall \ell \in \text{co}(v(X)).$$

Note: $\frac{\mathrm{d}}{\mathrm{d}\ell} \text{RHS} = \int \mathbf{1}_{k \leq \ell} F(\mathrm{d}k) = \int_{-\infty}^{\ell} \mathrm{d}F = F(\ell).$

Note: $\phi' \propto F \iff \exists \alpha > 0$ such that $\alpha \phi' = F.$

Sketch proof of Proposition 1

Note: (i) is equivalent (by change of var'ble $\ell := v(x)$) to:

$\exists \alpha > 0, \beta \in \mathbf{R} \ \& \ \phi : \text{co}(v(X)) \rightarrow \mathbf{R}$ such that
 $u = \phi \circ v$ and

$$\alpha \phi(\ell) + \beta = \int \max\{\ell, k\} F(dk) \quad \forall \ell \in \text{co}(v(X)).$$

Note: $\frac{d}{d\ell} \text{RHS} = \int \mathbf{1}_{k \leq \ell} F(dk) = \int_{-\infty}^{\ell} dF = F(\ell).$

Note: $\phi' \propto F \iff \exists \alpha > 0$ such that $\alpha \phi' = F.$

(i) implies $\phi' \propto F$: immediate.

Sketch proof of Proposition 1

Note: (i) is equivalent (by change of var'ble $\ell := v(x)$) to:

$\exists \alpha > 0, \beta \in \mathbf{R} \ \& \ \phi : \text{co}(v(X)) \rightarrow \mathbf{R}$ such that
 $u = \phi \circ v$ and

$$\alpha \phi(\ell) + \beta = \int \max\{\ell, k\} F(dk) \quad \forall \ell \in \text{co}(v(X)).$$

Note: $\frac{d}{d\ell} \text{RHS} = \int \mathbf{1}_{k \leq \ell} F(dk) = \int_{-\infty}^{\ell} dF = F(\ell).$

Note: $\phi' \propto F \iff \exists \alpha > 0$ such that $\alpha \phi' = F.$

$\phi' \propto F$ implies (i): if $\alpha \phi' = F \equiv \frac{d}{d\ell} \text{RHS}$ then

by fundamental th'm of calculus, $\exists \beta \in \mathbf{R}$ s.t. $\alpha \phi + \beta = \text{RHS}.$ ■

Plan

Setup and background

The outside-option model

How an o.o. shapes risk attitude (Proposition 1)

Identification (Theorem 1)

Comparative statics (Theorem 2)

What is special about o.o.? (Theorem 3)

Revealed risk attitude

Question: what can be learned about v from observing choice among lotteries?

Revealed risk attitude

Question: what can be learned about v from $\alpha u + \beta$?

Revealed risk attitude

Theorem 1. For $X \neq \emptyset$ and bounded-above $u, v : X \rightarrow \mathbf{R}$ with u Lipschitz w.r.t. v ,^{*} the following are equivalent:

- (a) u is less risk-averse than v .

Revealed risk attitude

Theorem 1. For $X \neq \emptyset$ and bounded-above $u, v : X \rightarrow \mathbf{R}$ with u Lipschitz w.r.t. v ,^{*} the following are equivalent:

- (a) u is less risk-averse than v .
- (b) $\exists \alpha > 0, \beta \in \mathbf{R}$ & an ECDF F such that

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(\mathrm{d}k) \quad \text{for every } x \in X$$

& $F > 0$ on $(\underline{\inf} v(X), +\infty)$.[†]

Revealed risk attitude

Theorem 1. For $X \neq \emptyset$ and bounded-above $u, v : X \rightarrow \mathbf{R}$ with u Lipschitz w.r.t. v ,^{*} the following are equivalent:

- (a) u is less risk-averse than v .
- (b) $\exists \alpha > 0, \beta \in \mathbf{R}$ & an ECDF F such that

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(\mathrm{d}k) \quad \text{for every } x \in X$$

$$\& F > 0 \text{ on } (\underline{\inf} v(X), +\infty).^\dagger$$

^{*} $\exists L \geq 0$ s.t. $\forall x, y \in X, |u(x) - u(y)| \leq L|v(x) - v(y)|.$

Revealed risk attitude

Theorem 1. For $X \neq \emptyset$ and bounded-above $u, v : X \rightarrow \mathbf{R}$ with u Lipschitz w.r.t. v ,^{*} the following are equivalent:

- (a) u is less risk-averse than v .
- (b) $\exists \alpha > 0, \beta \in \mathbf{R}$ & an ECDF F such that

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(\mathrm{d}k) \quad \text{for every } x \in X$$

& $F > 0$ on $(\underline{\inf} v(X), +\infty)$.[†]

^{*} $\exists L \geq 0$ s.t. $\forall x, y \in X, |u(x) - u(y)| \leq L|v(x) - v(y)|$.

[†]Ordinarily $\underline{\inf} A := \inf A$. The exception:

$\underline{\inf} A := \inf(A \setminus \{\inf A\})$ if $\inf A < \inf(A \setminus \{\inf A\}) \notin A$.

Sketch proof of Theorem 1

(b) implies u less risk-averse than v :

$$- \text{(b)} + \text{Prop'n 1} \implies u = \phi \circ v \quad \& \quad \phi' \propto F.$$

Sketch proof of Theorem 1

(b) implies u less risk-averse than v :

$$- \text{(b) + Prop'n 1} \implies u = \phi \circ v \quad \& \quad \phi' \propto F.$$

$$- F \text{ increasing} \implies \phi \text{ convex.}$$

Sketch proof of Theorem 1

(b) implies u less risk-averse than v :

- (b) + Prop'n 1 $\implies u = \phi \circ v$ & $\phi' \propto F$.
- F increasing $\implies \phi$ convex.
- So by Pratt's theorem, u less risk-averse than v .

Sketch proof of Theorem 1

(b) implies u less risk-averse than v :

- (b) + Prop'n 1 $\implies u = \phi \circ v$ & $\phi' \propto F$.
- F increasing $\implies \phi$ convex.
- So by Pratt's theorem, u less risk-averse than v .

u less risk-averse than v implies (b):

- By Prop'n 1, suffices to find an ECDF F such that $u = \phi \circ v$ & $\phi' \propto F$.

Sketch proof of Theorem 1

(b) implies u less risk-averse than v :

- (b) + Prop'n 1 $\implies u = \phi \circ v$ & $\phi' \propto F$.
- F increasing $\implies \phi$ convex.
- So by Pratt's theorem, u less risk-averse than v .

u less risk-averse than v implies (b):

- By Prop'n 1, suffices to find an ECDF F such that $u = \phi \circ v$ & $\phi' \propto F$.
- u LRA v + Pratt's th'm $\implies u = \phi \circ v$ & ϕ convex.

Sketch proof of Theorem 1

(b) implies u less risk-averse than v :

- (b) + Prop'n 1 $\implies u = \phi \circ v$ & $\phi' \propto F$.
- F increasing $\implies \phi$ convex.
- So by Pratt's theorem, u less risk-averse than v .

u less risk-averse than v implies (b):

- By Prop'n 1, suffices to find an ECDF F such that $u = \phi \circ v$ & $\phi' \propto F$.
- u LRA v + Pratt's th'm $\implies u = \phi \circ v$ & ϕ convex.
- Define $F := \phi' / \sup \phi'$. It's an ECDF. ■

Alternative proof that (a) implies (b)

Call $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ inflationary iff $\psi(\ell) \geq \ell \quad \forall \ell$.

Let $\mathcal{F}$ be set of all $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ that are increasing, convex, 1-Lipschitz & inflationary.

Alternative proof that (a) implies (b)

Call $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ inflationary iff $\psi(\ell) \geq \ell \quad \forall \ell$.

Let $\mathcal{F}$ be set of all $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ that are increasing, convex, 1-Lipschitz & inflationary.

$u \text{ LRA } v + \text{Pratt's th'm} \implies u = \phi \circ v \quad \& \quad \phi \text{ convex.}$

Alternative proof that (a) implies (b)

Call $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ inflationary iff $\psi(\ell) \geq \ell \quad \forall \ell$.

Let $\mathcal{F}$ be set of all $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ that are increasing, convex, 1-Lipschitz & inflationary.

$u \text{ LRA } v + \text{Pratt's th'm} \implies u = \phi \circ v \quad \& \quad \phi \text{ convex.}$

By regularity ass'ns, $\exists \alpha > 0 \quad \& \quad \beta \in \mathbf{R} \text{ s.t. } \alpha\phi + \beta \in \mathcal{F}$:

- Lipschitz ass'n $\implies \exists \alpha > 0 \text{ s.t. } \alpha\phi \text{ is 1-Lipschitz.}$
- v b'd'd above $\implies \exists \beta \in \mathbf{R} \text{ s.t. } \alpha\phi + \beta \text{ is infl'ary.}$

Alternative proof that (a) implies (b)

Call $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ inflationary iff $\psi(\ell) \geq \ell \quad \forall \ell$.

Let $\mathcal{F}$ be set of all $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ that are increasing, convex, 1-Lipschitz & inflationary.

$u \text{ LRA } v + \text{Pratt's th'm} \implies u = \phi \circ v \text{ \& } \phi \text{ convex.}$

By regularity ass'ns, $\exists \alpha > 0 \text{ \& } \beta \in \mathbf{R} \text{ s.t. } \alpha\phi + \beta \in \mathcal{F}$:

- Lipschitz ass'n $\implies \exists \alpha > 0 \text{ s.t. } \alpha\phi \text{ is 1-Lipschitz.}$
- v b'd'd above $\implies \exists \beta \in \mathbf{R} \text{ s.t. } \alpha\phi + \beta \text{ is infl'ary.}$

So by Choquet's theorem, $\alpha\phi + \beta$ equals a mixture of extreme points of $\mathcal{F}$: $\exists$ dist'n ν on $\text{extr}\mathcal{F}$ such that

$$\alpha\phi(\ell) + \beta = \int \psi(\ell) \nu(d\psi) \quad \text{for every } \ell \in \text{co}(v(X)).$$

Alternative proof that (a) implies (b)

Call $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ inflationary iff $\psi(\ell) \geq \ell \quad \forall \ell$.

Let $\mathcal{F}$ be set of all $\psi : \text{co}(v(X)) \rightarrow \mathbf{R}$ that are increasing, convex, 1-Lipschitz & inflationary.

$u \text{ LRA } v + \text{Pratt's th'm} \implies u = \phi \circ v \text{ \& } \phi \text{ convex.}$

By regularity ass'ns, $\exists \alpha > 0 \text{ \& } \beta \in \mathbf{R} \text{ s.t. } \alpha\phi + \beta \in \mathcal{F}$:

- Lipschitz ass'n $\implies \exists \alpha > 0 \text{ s.t. } \alpha\phi \text{ is 1-Lipschitz.}$
- v b'd'd above $\implies \exists \beta \in \mathbf{R} \text{ s.t. } \alpha\phi + \beta \text{ is infl'ary.}$

So by Choquet's theorem, $\alpha\phi + \beta$ equals a mixture of extreme points of $\mathcal{F}$: $\exists$ dist'n ν on $\text{extr}\mathcal{F}$ such that

$$\alpha\phi(\ell) + \beta = \int \psi(\ell) \nu(d\psi) \quad \text{for every } \ell \in \text{co}(v(X)).$$

Finally, $\text{extr}\mathcal{F} = \left\{ \ell \mapsto \max\{\ell, k\} : k \in [-\infty, +\infty) \right\}.$ ■

(Non-)identification of the o.o. dist'n

Question: what can be learned about F from observing choice among lotteries?

(Non-)identification of the o.o. dist'n

Question: what can be learned about F from $\alpha u + \beta$?

(Non-)identification of the o.o. dist'n

Question: what can be learned about F from $\alpha u + \beta$?

Answer: essentially nothing. (details: slide 33)

Plan

Setup and background

The outside-option model

How an o.o. shapes risk attitude (Proposition 1)

Identification (Theorem 1)

Comparative statics (Theorem 2)

What is special about o.o.? (Theorem 3)

Comparative statics

Question: how does effective risk att. u vary with (v, F) ?

Comparative statics

Theorem 2. Fix $X \neq \emptyset$,

$u, v : X \rightarrow \mathbf{R}$,

ECDF F with $F > 0$ on $(\inf v(X), +\infty)$.

Suppose $\exists \alpha > 0$ & $\beta \in \mathbf{R}$ such that

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(dk) \quad \forall x \in X.$$

Assume $(v(X) \text{ Borel and } F \text{ concentrated on } v(X) \cup \{-\infty\})$.

Comparative statics

Theorem 2. Fix $X \neq \emptyset$,

$\hat{u}, \hat{v} : X \rightarrow \mathbf{R}$,

ECDF $\hat{F}$ with $\hat{F} > 0$ on $(\inf \hat{v}(X), +\infty)$.

Suppose $\exists \hat{\alpha} > 0$ & $\hat{\beta} \in \mathbf{R}$ such that

$$\hat{\alpha}\hat{u}(x) + \hat{\beta} = \int \max\{\hat{v}(x), k\} \hat{F}(dk) \quad \forall x \in X.$$

Assume $(\hat{v}(X) \text{ Borel and}) \hat{F}$ concentrated on $\hat{v}(X) \cup \{-\infty\}$.

Comparative statics

Theorem 2. Fix $X \neq \emptyset$,

$\hat{u}, \hat{v} : X \rightarrow \mathbf{R}$,

ECDF $\hat{F}$ with $\hat{F} > 0$ on $(\inf \hat{v}(X), +\infty)$.

Suppose $\exists \hat{\alpha} > 0$ & $\hat{\beta} \in \mathbf{R}$ such that

$$\hat{\alpha} \hat{u}(x) + \hat{\beta} = \int \max\{\hat{v}(x), k\} \hat{F}(dk) \quad \forall x \in X.$$

Assume $(\hat{v}(X) \text{ Borel and}) \hat{F}$ concentrated on $\hat{v}(X) \cup \{-\infty\}$.

Comparative statics

Theorem 2. Fix $X \neq \emptyset$,

$\hat{u}, \hat{v} : X \rightarrow \mathbf{R}$,

ECDF $\hat{F}$ with $\hat{F} > 0$ on $(\inf \hat{v}(X), +\infty)$.

Suppose $\exists \hat{\alpha} > 0$ & $\hat{\beta} \in \mathbf{R}$ such that

$$\hat{\alpha}\hat{u}(x) + \hat{\beta} = \int \max\{\hat{v}(x), k\} \hat{F}(dk) \quad \forall x \in X.$$

Assume $(\hat{v}(X) \text{ Borel and}) \hat{F}$ concentrated on $\hat{v}(X) \cup \{-\infty\}$.

(a) Sps $\hat{v} = v$. Then $\hat{u}$ is less risk-averse than u iff $\hat{F}$ is better than F in the RHRO on $v(X) \setminus \{\sup v(X)\}$.

$\hookrightarrow$ RHRO (reverse hazard rate order) on $K \subseteq \mathbf{R}$:

$$F(\ell)\hat{F}(k) \leq F(k)\hat{F}(\ell) \text{ for all } k < \ell \text{ in } K.$$

Comparative statics

Theorem 2. Fix $X \neq \emptyset$,

$\hat{u}, \hat{v} : X \rightarrow \mathbf{R}$,

ECDF $\hat{F}$ with $\hat{F} > 0$ on $(\inf \hat{v}(X), +\infty)$.

Suppose $\exists \hat{\alpha} > 0$ & $\hat{\beta} \in \mathbf{R}$ such that

$$\hat{\alpha}\hat{u}(x) + \hat{\beta} = \int \max\{\hat{v}(x), k\} \hat{F}(dk) \quad \forall x \in X.$$

Assume $(\hat{v}(X) \text{ Borel and}) \hat{F}$ concentrated on $\hat{v}(X) \cup \{-\infty\}$.

(a) Sps $\hat{v} = v$. Then $\hat{u}$ is less risk-averse than u iff
 $\hat{F}$ is better than F in the RHRO on $v(X) \setminus \{\sup v(X)\}$.

(b) Sps $F \circ v = \hat{F} \circ \hat{v}$. Then $\hat{u}$ is less risk-averse than u iff
 $\hat{v}$ is less risk-averse than v .

$\hookrightarrow F \circ v = \hat{F} \circ \hat{v}$ means utility units change $(v \rightsquigarrow \hat{v})$,
 physical o.o. dist'n $\mu \in \Delta(X \cup \{\emptyset\})$ held fixed. (recall
sl. 12)

Comparative statics

Theorem 2. Fix $X \neq \emptyset$,

$\hat{u}, \hat{v} : X \rightarrow \mathbf{R}$,

ECDF $\hat{F}$ with $\hat{F} > 0$ on $(\inf \hat{v}(X), +\infty)$.

Suppose $\exists \hat{\alpha} > 0$ & $\hat{\beta} \in \mathbf{R}$ such that

$$\hat{\alpha}\hat{u}(x) + \hat{\beta} = \int \max\{\hat{v}(x), k\} \hat{F}(dk) \quad \forall x \in X.$$

Assume $(\hat{v}(X) \text{ Borel and}) \hat{F}$ concentrated on $\hat{v}(X) \cup \{-\infty\}$.

(a) Sps $\hat{v} = v$. Then $\hat{u}$ is less risk-averse than u iff
 $\hat{F}$ is better than F in the RHRO on $v(X) \setminus \{\sup v(X)\}$.

(b) Sps $F \circ v = \hat{F} \circ \hat{v}$. Then $\hat{u}$ is less risk-averse than u iff
 $\hat{v}$ is less risk-averse than v .

Comparative statics: monetary case

Theorem 2. Fix $X \neq \emptyset$ open convex subset of $\mathbf{R}$,
 $\hat{u}, \hat{v} : X \rightarrow \mathbf{R}$ that are str. incr.,
 ECDF $\hat{G}$ with $\hat{G} > 0$ on $(\inf X, +\infty)$.
 Suppose $\exists \hat{\alpha} > 0$ & $\hat{\beta} \in \mathbf{R}$ such that

$$\hat{\alpha}\hat{u}(x) + \hat{\beta} = \int \max\{\hat{v}(x), \hat{v}(y)\} \hat{G}(dy) \quad \forall x \in X.$$

Assume $\hat{G}$ concentrated on $X \cup \{-\infty\}$.

(a) Sps $\hat{v} = v$. Then $\hat{u}$ is less risk-averse than u iff
 $\hat{G}$ is better than G in the RHRO on $X \setminus \{\sup X\}$.

$\hookrightarrow$ RHRO (reverse hazard rate order) on $Y \subseteq \mathbf{R}$:
 $G(y)\hat{G}(x) \leq G(x)\hat{G}(y)$ for all $x < y$ in Y .

(b) Sps $G = \hat{G}$. Then $\hat{u}$ is less risk-averse than u iff
 $\hat{v}$ is less risk-averse than v .

Comparative statics: smooth monetary case

By Prop'n 1,
$$\frac{u''}{u'} = \frac{v''}{v'} + \frac{G'}{G}.$$

Comparative statics: smooth monetary case

By Prop'n 1, $\frac{u''}{u'} = \frac{v''}{v'} + \frac{G'}{G}.$

When v is held fixed,

$$\underbrace{u \text{ becomes less risk-averse}}_{\frac{u''}{u'} \nearrow \text{ pointwise}} \text{ iff } \underbrace{G \text{ improves in RHRO.}}_{\frac{G'}{G} \nearrow \text{ pointwise}}$$

Comparative statics: smooth monetary case

By Prop'n 1, $\frac{u''}{u'} = \frac{v''}{v'} + \frac{G'}{G}.$

When v is held fixed,

$$\underbrace{u \text{ becomes less risk-averse}}_{\frac{u''}{u'} \nearrow \text{pointwise}} \text{ iff } \underbrace{G \text{ improves in RHRO.}}_{\frac{G'}{G} \nearrow \text{pointwise}}$$

When G is held fixed,

$$\underbrace{u \text{ becomes less risk-averse}}_{\frac{u''}{u'} \nearrow \text{pointwise}} \text{ iff } \underbrace{v \text{ becomes less risk-averse.}}_{\frac{v''}{v'} \nearrow \text{pointwise}} \blacksquare$$

Plan

Setup and background

The outside-option model

How an o.o. shapes risk attitude (Proposition 1)

Identification (Theorem 1)

Comparative statics (Theorem 2)

What is special about o.o.? (Theorem 3)

Idea

Adding an o.o. to a decision problem transforms it
by replacing each $x \in X$ with an x -contingent lottery.
(Viz. lottery which returns x or the o.o., whichever is better.)

Idea

Adding an o.o. to a decision problem transforms it
by replacing each $x \in X$ with an x -contingent lottery.
(Viz. lottery which returns x or the o.o., whichever is better.)

Another example of such transform'n: adding background risk.

Idea

Adding an o.o. to a decision problem transforms it by replacing each $x \in X$ with an x -contingent lottery. (Viz. lottery which returns x or the o.o., whichever is better.)

Another example of such transform'n: adding background risk.

Question: which transformations decrease risk-aversion?

Idea

Adding an o.o. to a decision problem transforms it by replacing each $x \in X$ with an x -contingent lottery. (Viz. lottery which returns x or the o.o., whichever is better.)

Another example of such transform'n: adding background risk.

Question: which transformations decrease risk-aversion?

- adding o.o.: yes, by Th'm 1.
- adding background risk: no, not in general.

Idea

Adding an o.o. to a decision problem transforms it by replacing each $x \in X$ with an x -contingent lottery. (Viz. lottery which returns x or the o.o., whichever is better.)

Another example of such transform'n: adding background risk.

Question: which transformations decrease risk-aversion?

Answer: only transform'ns that amount to adding an o.o.

The question, formally

Transform'n: replace each x with draw from measure μ_x on X .

'True' risk att. $v : X \rightarrow \mathbf{R}$ $\rightsquigarrow$ effective risk att. $x \mapsto \int v d\mu_x$.

The question, formally

Transform'n: replace each x with draw from measure μ_x on X .

'True' risk att. $v : X \rightarrow \mathbf{R}$ $\rightsquigarrow$ effective risk att. $x \mapsto \int v d\mu_x$.

Question: given rich class $\mathcal{V}$ of possible 'true' risk attitudes,
which families $(\mu_x)_{x \in X}$ have the property that
 $\forall v \in \mathcal{V}, \quad x \mapsto \int v d\mu_x$ is less risk-averse than v ?

The question, reformulated

Question: given rich class $\mathcal{V}$ of possible ‘true’ risk attitudes, which families $(\mu_x)_{x \in X}$ have the property that $\forall v \in \mathcal{V}, x \mapsto \int v d\mu_x$ is less risk-averse than v ?

Since u less risk-averse than $v \implies u, v$ ordinally equivalent, restrict to $\mathcal{V}$ s whose members are mutually ordinally equivalent.

The question, reformulated

Question: given rich class $\mathcal{V}$ of possible ‘true’ risk attitudes, which families $(\mu_x)_{x \in X}$ have the property that $\forall v \in \mathcal{V}, \quad x \mapsto \int v d\mu_x$ is less risk-averse than v ?

Since u less risk-averse than $v \implies u, v$ ordinally equivalent, restrict to $\mathcal{V}$ s whose members are mutually ordinally equivalent.

No other restrictions $\implies \mathcal{V}$ is an ordinal equivalence class.

The question, reformulated

Question: given rich class $\mathcal{V}$ of possible ‘true’ risk attitudes, which families $(\mu_x)_{x \in X}$ have the property that $\forall v \in \mathcal{V}, \quad x \mapsto \int v d\mu_x$ is less risk-averse than v ?

Since u less risk-averse than $v \implies u, v$ ordinally equivalent, restrict to $\mathcal{V}$ s whose members are mutually ordinally equivalent.

No other restrictions $\implies \mathcal{V}$ is an ordinal equivalence class.

Given such $\mathcal{V}$, fix $v_0 \in \mathcal{V}$ & embed X in $\mathbf{R}$ via $x \mapsto v_0(x)$. Then $X \subseteq \mathbf{R}$ & $\mathcal{V}$ is the set of all str. incr. f’ns $X \rightarrow \mathbf{R}$.

The question, reformulated

Question: given rich class $\mathcal{V}$ of possible ‘true’ risk attitudes, which families $(\mu_x)_{x \in X}$ have the property that $\forall v \in \mathcal{V}, \quad x \mapsto \int v d\mu_x$ is less risk-averse than v ?

Since u less risk-averse than $v \implies u, v$ ordinally equivalent, restrict to $\mathcal{V}$ s whose members are mutually ordinally equivalent.

No other restrictions $\implies \mathcal{V}$ is an ordinal equivalence class.

Given such $\mathcal{V}$, fix $v_0 \in \mathcal{V}$ & embed X in $\mathbf{R}$ via $x \mapsto v_0(x)$. Then $X \subseteq \mathbf{R}$ & $\mathcal{V}$ is the set of all str. incr. f’ns $X \rightarrow \mathbf{R}$.

Question’: which families $(G_x)_{x \in X}$ of CDFs concentr’d on X have the property that $\forall$ str. incr. $v : X \rightarrow \mathbf{R}$, $x \mapsto \int v dG_x$ is less risk-averse than v ?

The answer

Theorem 3. For non-empty Borel $X \subseteq \mathbf{R}$ with $\sup X \notin X$ & family $(G_x)_{x \in X}$ of CDFs concentrated on X , the following are equivalent:

- (a) $\forall$ bounded str. incr. $v : X \rightarrow \mathbf{R}$,
 $x \mapsto \int v dG_x$ is less risk-averse than v .

The answer

Theorem 3. For non-empty Borel $X \subseteq \mathbf{R}$ with $\sup X \notin X$ & family $(G_x)_{x \in X}$ of CDFs concentrated on X , the following are equivalent:

- (a) $\forall$ bounded str. incr. $v : X \rightarrow \mathbf{R}$,
 $x \mapsto \int v dG_x$ is less risk-averse than v .
- (b) $\exists$ ECDF $G \left\{ \begin{array}{l} \text{concentrated on } X \cup \{-\infty\}, \\ \text{with } G > 0 \text{ on } (\inf X, +\infty) \end{array} \right\}$ such that

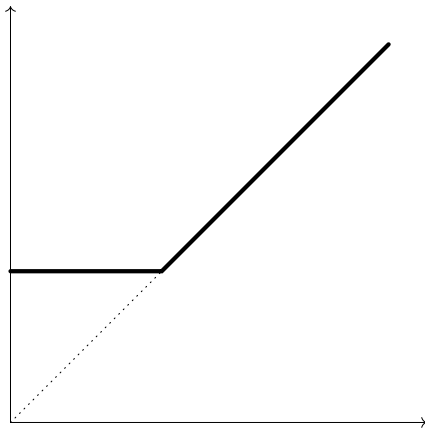
$\forall$ bounded str. incr. $v : X \rightarrow \mathbf{R}$,

$\exists \alpha > 0$ & $\beta \in \mathbf{R}$ such that

$$\alpha \int v dG_x + \beta = \int \max\{v(x), v(y)\} G(dy) \quad \text{for every } x \in X,$$

where by convention $v(-\infty) := -\infty$.

Thanks!



Adam Smith on limited liability and risk-taking

From the *Wealth of Nations*.³

In a private copartnery, each partner is bound for the debts contracted by the company to the whole extent of his fortune. In a joint stock company, on the contrary, each partner is bound only to the extent of his share. This total exemption from trouble and from risk, beyond a limited sum, encourages many people to become adventurers in joint stock companies, who would, upon no account, hazard their fortunes in any private copartnery.

(back to slide 4)

³Pages. 740–1 in the Glasgow edition (Smith, 1776/1976).

Extended CDFs

Extended CDF: $k \mapsto \mathbf{P}(K \leq k)$ for a $[-\infty, +\infty)$ -valued random variable K .

Observation: $F : \mathbf{R} \rightarrow [0, 1]$ is an ECDF iff
increasing, right-continuous, $\lim_{k \nearrow +\infty} F(k) = 1$.

Expectation of meas'ble $g : [-\infty, +\infty) \rightarrow \mathbf{R}$ w.r.t. ECDF F :

$$\int g dF := \left(\lim_{k \searrow -\infty} F(k) \right) g(-\infty) + \int_{\mathbf{R}} g dF.$$

(back to slide 11)

(Non-)identification of the o.o. dist'n: details

Proposition 2. For $X \neq \emptyset$, $u : X \rightarrow \mathbf{R}$ & ECDF F , the following are equivalent:

- (I) $\int_{(0,+\infty)} kF(dk) < +\infty$, and
if $\int_{(-\infty,0]} kF(dk) > -\infty$ then u is bounded below.

(Non-)identification of the o.o. dist'n: details

Proposition 2. For $X \neq \emptyset$, $u : X \rightarrow \mathbf{R}$ & ECDF F , the following are equivalent:

(I) $\int_{(0,+\infty)} kF(dk) < +\infty$, and
if $\int_{(-\infty,0]} kF(dk) > -\infty$ then u is bounded below.

(II) $\exists \alpha > 0$, $\beta \in \mathbf{R}$, & $v : X \rightarrow \mathbf{R}$ such that

$$\alpha u(x) + \beta = \int \max\{v(x), k\} F(dk) \quad \forall x \in X.$$

(back to slide 21)

Forces shaping risk att.: background risk

E.g. Ross (1981), Kihlstrom, Romer and Williams (1981), Pratt and Zeckhauser (1987), Kimball (1993), Eeckhoudt, Gollier and Schlesinger (1996), Gollier and Pratt (1996), Pomatto, Strack and Tamuz (2020) and Mu, Pomatto, Strack and Tamuz (2024).

(back to slide 5)

Forces shaping risk att.: employment contracts

E.g. Wilson (1969), Ross (1974), Amihud and Lev (1981), Lambert (1986), Hirshleifer and Suh (1992), Diamond (1998), Garicano and Rayo (2016) and Barron, Georgiadis and Swinkels (2020).

(back to slide 5)

Forces shaping risk att.: financing contracts

E.g. Galai and Masulis (1976), Jensen and Meckling (1976), Stiglitz and Weiss (1981), Green (1984) and Hébert (2018).

(back to slide 5)

Forces shaping risk att.: career concerns

E.g. Holmström (1982/1999), Hirshleifer and Thakor (1992), Hermalin (1993) and Chen (2015).

(back to slide 5)

Forces shaping risk att.: flexibility

E.g. Drèze and Modigliani (1966/1972), Mossin (1969), Spence and Zeckhauser (1972), Machina (1982, 1984), Bodie, Merton and Samuelson (1992), Gollier (2005), Chetty and Szeidl (2007) and Postlewaite, Samuelson and Silverman (2008).

(back to slide 5)

Forces shaping risk att.: R&D races

E.g. Dasgupta and Stiglitz (1980), Klette and de Meza (1986), Dasgupta and Maskin (1987), Cabral (2003) and Anderson and Cabral (2007).

(back to slide 5)

Forces shaping risk att.: competition for status

E.g. Robson (1992), Rosen (1997), Becker, Murphy and Werning (2005), Ray and Robson (2012) and Hopkins (2018).

(back to slide 5)

Forces shaping risk att.: competition generally

E.g. Hvide (2002), Hvide and Kristiansen (2003), Taylor (2003), Kräkel and Sliwka (2004), Seel and Strack (2013, 2016), Nutz and Zhang (2022) and Fang et al. (2025).

(back to slide 5)

Forces shaping risk att.: competition in banking

Surveys: Beck (2008), Carletti (2008), Vives (2016) and Berger, Klapper and Turk-Ariss (2017).

(back to slide 5)

References I

- Amihud, Y., & Lev, B. (1981). Risk reduction as a managerial motive for conglomerate mergers. *Bell Journal of Economics*, 12(2), 605–617.
<https://doi.org/10.2307/3003575>
- Anderson, A., & Cabral, L. M. B. (2007). Go for broke or play it safe? Dynamic competition with choice of variance. *RAND Journal of Economics*, 38(3), 593–609.
<https://doi.org/10.1111/j.0741-6261.2007.00102.x>
- Barron, D., Georgiadis, G., & Swinkels, J. (2020). Optimal contracts with a risk-taking agent. *Theoretical Economics*, 15(2), 715–761. <https://doi.org/10.3982/TE3660>
- Beck, T. (2008). *Bank competition and financial stability: Friends or foes?* [working paper, Jun 2008].

References II

- Becker, G. S., Murphy, K. M., & Werning, I. (2005). The equilibrium distribution of income and the market for status. *Journal of Political Economy*, 113(2), 282–310. <https://doi.org/10.1086/427463>
- Ben-Porath, E., Dekel, E., & Lipman, B. L. (2018). Disclosure and choice. *Review of Economic Studies*, 85(3), 1471–1501. <https://doi.org/10.1093/restud/rdx064>
- Berger, A. N., Klapper, L. F., & Turk-Ariss, R. (2017). Bank competition and financial stability. In J. A. Bikker & L. Spierdijk (Eds.), *Handbook of competition in banking and finance* (pp. 185–204). Elgar. <https://doi.org/10.4337/9781785363306.00018>
- Bodie, Z., Merton, R. C., & Samuelson, W. F. (1992). Labor supply flexibility and portfolio choice in a life cycle model. *Journal of Economic Dynamics and Control*, 16(3–4), 427–449. [https://doi.org/10.1016/0165-1889\(92\)90044-F](https://doi.org/10.1016/0165-1889(92)90044-F)

References III

- Cabral, L. M. B. (2003). R&D competition when firms choose variance. *Journal of Economics & Management Strategy*, 12(1), 139–150.
<https://doi.org/10.1111/j.1430-9134.2003.00139.x>
- Carletti, E. (2008). Competition and regulation in banking. In A. V. Thakor & A. W. A. Boot (Eds.), *Handbook of financial intermediation and banking* (pp. 449–482). Elsevier.
<https://doi.org/10.1016/B978-044451558-2.50022-2>
- Chen, Y. (2015). Career concerns and excessive risk taking. *Journal of Economics & Management Strategy*, 24(1), 110–130. <https://doi.org/10.1111/j.1430-9134.2015.00139.x>
- Chetty, R., & Szeidl, A. (2007). Consumption commitments and risk preferences. *Quarterly Journal of Economics*, 122(2), 831–877. <https://doi.org/10.1162/qjec.122.2.831>

References IV

- Dasgupta, P., & Maskin, E. (1987). The simple economics of research portfolios. *The Economic Journal*, 97(387), 581–595. <https://doi.org/10.2307/2232925>
- Dasgupta, P., & Stiglitz, J. (1980). Uncertainty, industrial structure, and the speed of r&d. 11(1), 1–28. <https://doi.org/10.2307/3003398>
- Diamond, P. (1998). Managerial incentives: On the near linearity of optimal compensation. *Journal of Political Economy*, 106(5), 931–957. <https://doi.org/10.1086/250036>
- Drèze, J. H., & Modigliani, F. (1966). Épargne et consommation en avenir aléatoire. *Cahiers du Séminaire d'Économétrie*, 9, 7–33.
- Drèze, J. H., & Modigliani, F. (1972). Consumption decisions under uncertainty. *Journal of Economic Theory*, 5(3), 308–335. [https://doi.org/10.1016/0022-0531\(72\)90044-0](https://doi.org/10.1016/0022-0531(72)90044-0)

References V

- Eeckhoudt, L., Gollier, C., & Schlesinger, H. (1996). Changes in background risk and risk taking behavior. *Econometrica*, 64(3), 683–689. <https://doi.org/10.2307/2171866>
- Fang, D., Ke, C., Kubitz, G., Noe, T. H., Page, L., & Liu, Y. (2025). *Winning ways: How rank-based incentives shape risk-taking decisions* [working paper, 5 Feb 2025]. <https://doi.org/10.2139/ssrn.4856341>
- Galai, D., & Masulis, R. W. (1976). The option pricing model and the risk factor of stock. *Journal of Financial Economics*, 3(1–2), 53–81. [https://doi.org/10.1016/0304-405X\(76\)90020-9](https://doi.org/10.1016/0304-405X(76)90020-9)
- Garicano, L., & Rayo, L. (2016). Why organizations fail: Models and cases. *Journal of Economic Literature*, 54(1), 137–192. <https://doi.org/10.1257/jel.54.1.137>

References VI

- Golbe, D. L. (1981). The effects of imminent bankruptcy on stockholder risk preferences and behavior. *Bell Journal of Economics*, 12(1), 321–328.
<https://doi.org/10.2307/3003528>
- Golbe, D. L. (1988). Risk-taking by firms near bankruptcy. *Economics Letters*, 28(1), 75–79.
[https://doi.org/10.1016/0165-1765\(88\)90075-4](https://doi.org/10.1016/0165-1765(88)90075-4)
- Gollier, C. (2005). *Does flexibility enhance risk tolerance?* [working paper, 8 Jun 2025].
- Gollier, C., Koehl, P.-F., & Rochet, J.-C. (1997). Risk-taking behavior with limited liability and risk aversion. *Journal of Risk and Insurance*, 64(2), 347–370.
<https://doi.org/10.2307/253734>
- Gollier, C., & Pratt, J. W. (1996). Risk vulnerability and the tempering effect of background risk. *Econometrica*, 64(5), 1109–1123. <https://doi.org/10.2307/2171958>

References VII

- Green, R. C. (1984). Investment incentives, debt, and warrants. *Journal of Financial Economics*, 13(1), 115–136.
[https://doi.org/10.1016/0304-405X\(84\)90034-5](https://doi.org/10.1016/0304-405X(84)90034-5)
- Hébert, B. (2018). Moral hazard and the optimality of debt. *Review of Economic Studies*, 85(4), 2214–2252.
<https://doi.org/10.1093/restud/rdx080>
- Hermalin, B. E. (1993). Managerial preferences concerning risky projects. *Journal of Law, Economics, & Organization*, 9(1), 127–135.
<https://doi.org/10.1093/oxfordjournals.jleo.a037049>
- Hirshleifer, D., & Suh, Y. (1992). Risk, managerial effort, and project choice. *Journal of Financial Intermediation*, 2(3), 308–345. [https://doi.org/10.1016/1042-9573\(92\)90004-W](https://doi.org/10.1016/1042-9573(92)90004-W)
- Hirshleifer, D., & Thakor, A. V. (1992). Managerial conservatism, project choice, and debt. *Review of Financial Studies*, 5(3), 437–470. <https://doi.org/10.1093/rfs/5.3.437>

References VIII

- Holmström, B. (1982). Managerial incentive problems: A dynamic perspective [reprinted in Holmström (1999)]. In *Vetenskap och företagsledning: Studier i ekonomi och ledarskap tillägnade Lars Wahlbeck* (pp. 209–230). Swedish School of Economics and Business Administration, Helsinki.
- Holmström, B. (1999). Managerial incentive problems: A dynamic perspective [reprinted from Holmström (1982)]. *Review of Economic Studies*, 66(1), 169–182.
<https://doi.org/10.1111/1467-937X.00083>
- Hopkins, E. (2018). Inequality and risk-taking behaviour. *Games and Economic Behavior*, 107, 316–328.
<https://doi.org/10.1016/j.geb.2017.11.007>
- Hvide, H. K. (2002). Tournament rewards and risk taking. *Journal of Labor Economics*, 20(4), 709–978.
<https://doi.org/10.1086/342041>

References IX

- Hvide, H. K., & Kristiansen, E. G. (2003). Risk taking in selection contests. *Games and Economic Behavior*, 42(1), 172–179. [https://doi.org/10.1016/S0899-8256\(02\)00538-9](https://doi.org/10.1016/S0899-8256(02)00538-9)
- Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4), 305–360. [https://doi.org/10.1016/0304-405X\(76\)90026-X](https://doi.org/10.1016/0304-405X(76)90026-X)
- Kihlstrom, R. E., Romer, D., & Williams, S. (1981). Risk aversion with random initial wealth. *Econometrica*, 49(4), 911–920. <https://doi.org/10.2307/1912510>
- Kimball, M. S. (1993). Standard risk aversion. *Econometrica*, 61(3), 589–611. <https://doi.org/10.2307/2951719>
- Klette, T., & de Meza, D. (1986). Is the market biased against risky R&D? *RAND Journal of Economics*, 17(1), 133–139.

References X

- Kräkel, M., & Sliwka, D. (2004). Risk taking in asymmetric tournaments. *German Economic Review*, 5(1), 103–116.
<https://doi.org/10.1111/j.1465-6485.2004.00096.x>
- Lambert, R. A. (1986). Executive effort and selection of risky projects. *RAND Journal of Economics*, 17(1), 77–88.
- Machina, M. J. (1982). Flexibility and the demand for risky assets. *Economics Letters*, 10(1–2), 71–76.
[https://doi.org/10.1016/0165-1765\(82\)90118-5](https://doi.org/10.1016/0165-1765(82)90118-5)
- Machina, M. J. (1984). Temporal risk and the nature of induced preferences. *Journal of Economic Theory*, 33(2), 199–231.
[https://doi.org/10.1016/0022-0531\(84\)90087-5](https://doi.org/10.1016/0022-0531(84)90087-5)
- Mossin, J. (1969). A note on uncertainty and preferences in a temporal context. *American Economic Review*, 59(1), 172–174.

References XI

- Mu, X., Pomatto, L., Strack, P., & Tamuz, O. (2024).
Background risk and small-stakes risk aversion. *American Economic Review: Insights*, 6(2), 262–276.
<https://doi.org/10.1257/aeri.20220480>
- Nutz, M., & Zhang, Y. (2022). Reward design in risk-taking contests. *SIAM Journal on Financial Mathematics*, 13(1), 129–146. <https://doi.org/10.1137/21M1397386>
- Pomatto, L., Strack, P., & Tamuz, O. (2020). Stochastic dominance under independent noise. *Journal of Political Economy*, 128(5), 1877–1900.
<https://doi.org/10.1086/705555>
- Postlewaite, A., Samuelson, L., & Silverman, D. (2008).
Consumption commitments and employment contracts. *Review of Economic Studies*, 75(2), 559–578.
<https://doi.org/10.1111/j.1467-937X.2008.00472.x>

References XII

- Pratt, J. W., & Zeckhauser, R. J. (1987). Proper risk aversion. *Econometrica*, 55(1), 143–154.
<https://doi.org/10.2307/1911160>
- Ray, D., & Robson, A. J. (2012). Status, intertemporal choice, and risk-taking. *Econometrica*, 80(4), 1505–1531.
<https://doi.org/10.3982/ECTA9042>
- Robson, A. J. (1992). Status, the distribution of wealth, private and social attitudes to risk. *Econometrica*, 60(4), 837–857. <https://doi.org/10.2307/2951568>
- Rosen, S. (1997). Manufactured inequality. *Journal of Labor Economics*, 15(2), 189–196.
<https://doi.org/10.1086/209829>
- Ross, S. A. (1974). On the economic theory of agency and the principle of similarity. In M. Balch, D. McFadden, & S.-Y. Wu (Eds.), *Essays on economic behavior under uncertainty* (pp. 215–237). North-Holland.

References XIII

- Ross, S. A. (1981). Some stronger measures of risk aversion in the small and the large with applications. *Econometrica*, 49(3), 621–638. <https://doi.org/10.2307/1911515>
- Seel, C., & Strack, P. (2013). Gambling in contests. *Journal of Economic Theory*, 148(5), 2033–2048. <https://doi.org/10.1016/j.jet.2013.07.005>
- Seel, C., & Strack, P. (2016). Continuous time contests with private information. *Mathematics of Operations Research*, 41(3), 745–1160. <https://doi.org/10.1287/moor.2015.0769>
- Smith, A. (1776). *An inquiry into the nature and causes of the wealth of nations*. W. Strahan; T. Cadell.
- Smith, A. (1776/1976). *An inquiry into the nature and causes of the wealth of nations* (W. B. Todd, Ed.; Vol. 2). Oxford University Press. <https://doi.org/10.1093/actrade/9780199269570.book.1>

References XIV

- Spence, M., & Zeckhauser, R. (1972). The effect of the timing of consumption decisions and the resolution of lotteries on the choice of lotteries. *Econometrica*, 40(2), 401–403.
<https://doi.org/10.2307/1909418>
- Stiglitz, J. E., & Weiss, A. (1981). Credit rationing in markets with imperfect information. *American Economic Review*, 71(3), 393–410.
- Taylor, J. (2003). Risk-taking behavior in mutual fund tournaments. *Journal of Economic Behavior & Organization*, 50(3), 373–383.
[https://doi.org/10.1016/S0167-2681\(02\)00028-8](https://doi.org/10.1016/S0167-2681(02)00028-8)
- Vives, X. (2016). *Competition and stability in banking: The role of regulation and competition policy*. Princeton University Press. <https://doi.org/10.23943/princeton/9780691171791.001.0001>

References XV

Wilson, R. (1969). The structure of incentives for decentralization under uncertainty. In *La décision: Agrégation et dynamique des ordres de préférence* (pp. 287–307). Éditions du CNRS.